Adaptive fractional dynamics in the visual cortex

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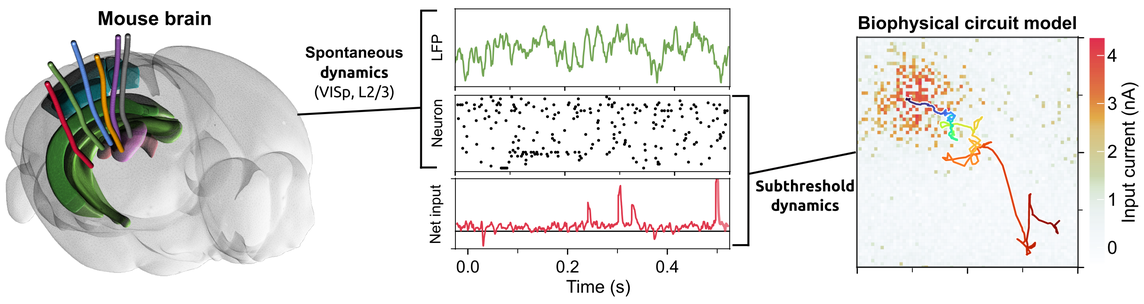
1 | What is the working regime of the visual cortex?

Fluctuation-driven regime

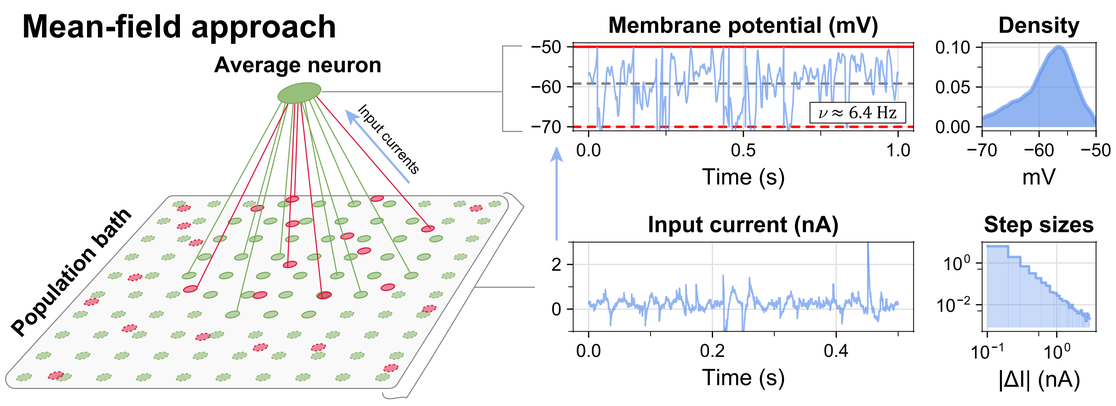
- Low firing rates [1]
- Sensitivity & sparse coding
- Spiking from fluctuations about low mean voltage [2]

Complex dynamics

- Nested oscillations and $1/f$ spectral components [3]
- Super-Poissonian spikes [4]
- Bumpy synaptic inputs [5]

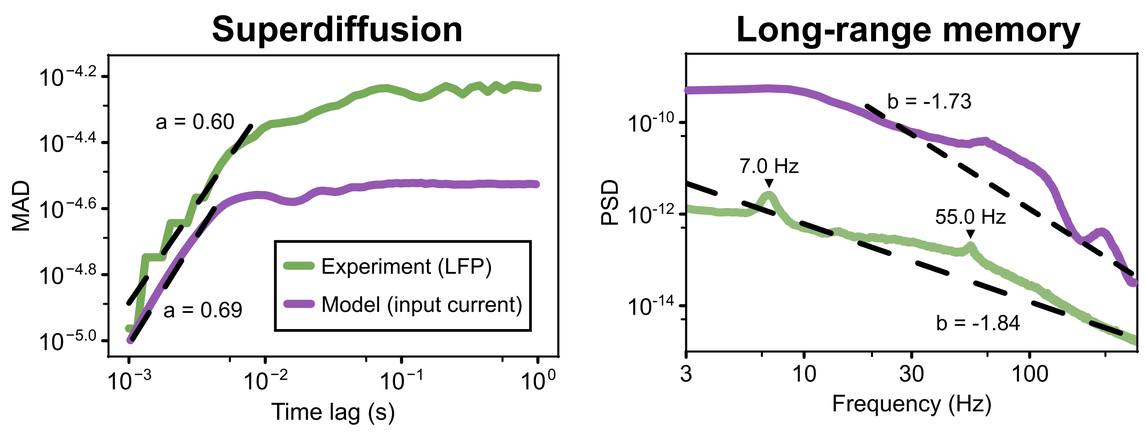


Mean-field theory of classical E-I networks captures the **fluctuation-driven regime**, but struggles to explain **complex dynamics** and **variability**.



Complex cortical dynamics are not consistent with the **normal diffusion** of the classical mean field [6]. Instead, they exhibit **anomalous diffusion**.

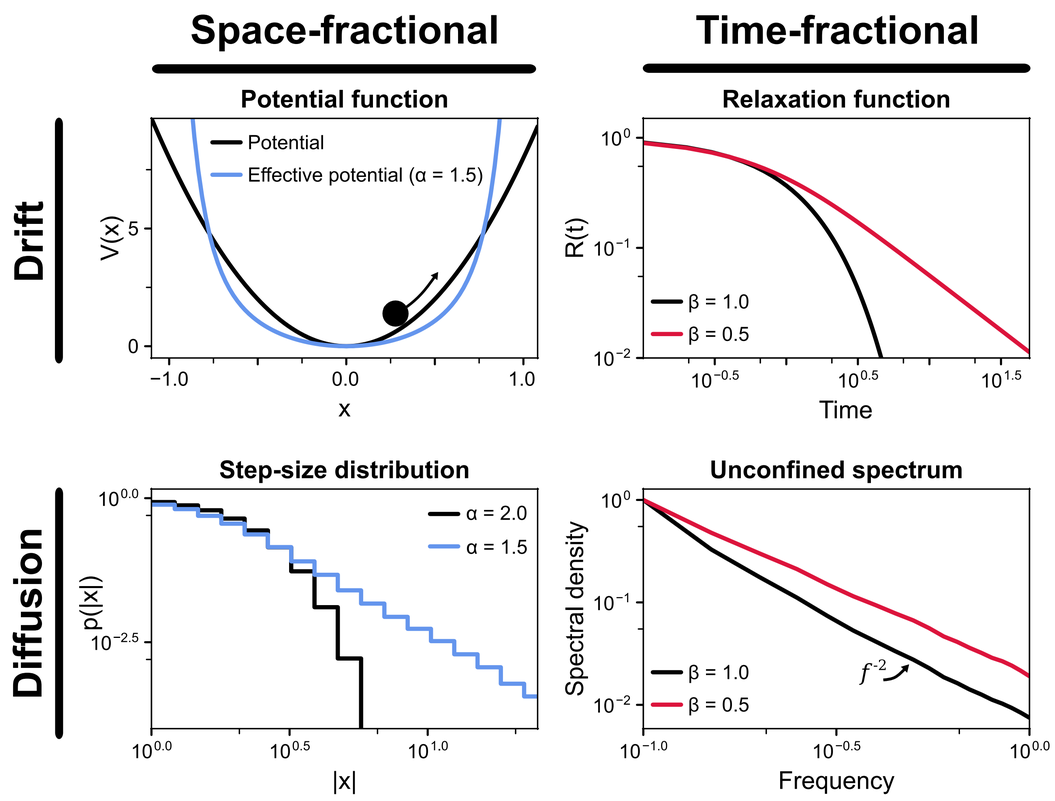
Classical mean field	Mouse cortex + model
Diffusion exponent $a = 0.5$	Diffusion exponent $a > 0.5$
⇒ Brownian motion	⇒ Superdiffusion
Spectral exponent $b = -2$	Spectral exponent $b > -2$
⇒ Memoryless inputs	⇒ Long-range memory



2 | How do we capture anomalous dynamics?

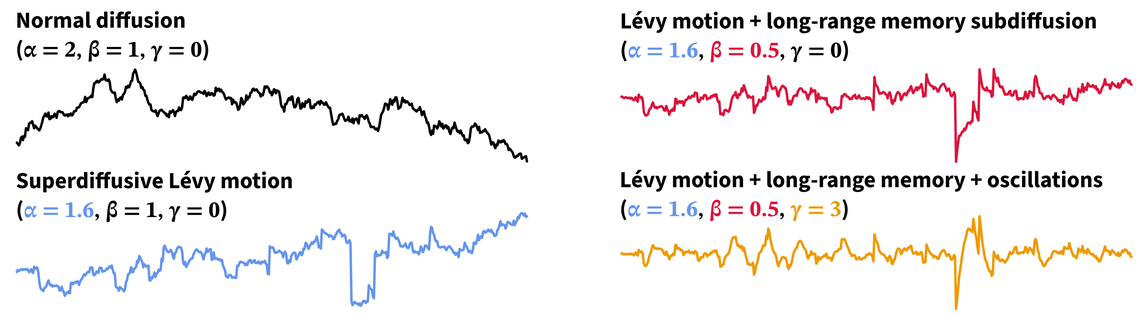
We generalize classical mean-field theory using **fractional calculus**:

- **Space-fractional**: power-law step-size distribution drives superdiffusion
- **Time-fractional**: power-law memory kernel adds long-range memory



3 | Bi-fractional neural sampling model

$$ {}^{C}D_t^{\beta} x = -\eta \nabla \tilde{V}_{\alpha} + \gamma p + \eta^{\frac{1}{\alpha}} \xi_{\alpha,\beta}\,, \quad \frac{dp}{dt} = -\gamma \nabla \tilde{V}_{\alpha} $$



$\alpha \in (1, 2]$: **spatial** fractional order

$\beta \in (0, 1]$: **temporal** fractional order

γ: **momentum coupling** (→ oscillations)

η: noise intensity/diffusion timescale

$\xi_{\alpha,\beta}$: linear fractional stable noise with $H = 1/2 + 1/\alpha - \beta/2$

Fractional spatial derivative [7]

$$ \nabla \tilde{V}_{\alpha}(x) = \frac{-\nabla\left[\mathcal{D}_x^{\alpha-2}\pi\right]}{\pi} $$

$$ \widehat{\mathcal{D}_x^{\alpha} f}(k) = -|k|^{\alpha} \hat{f}(k) $$

Fractional temporal derivative

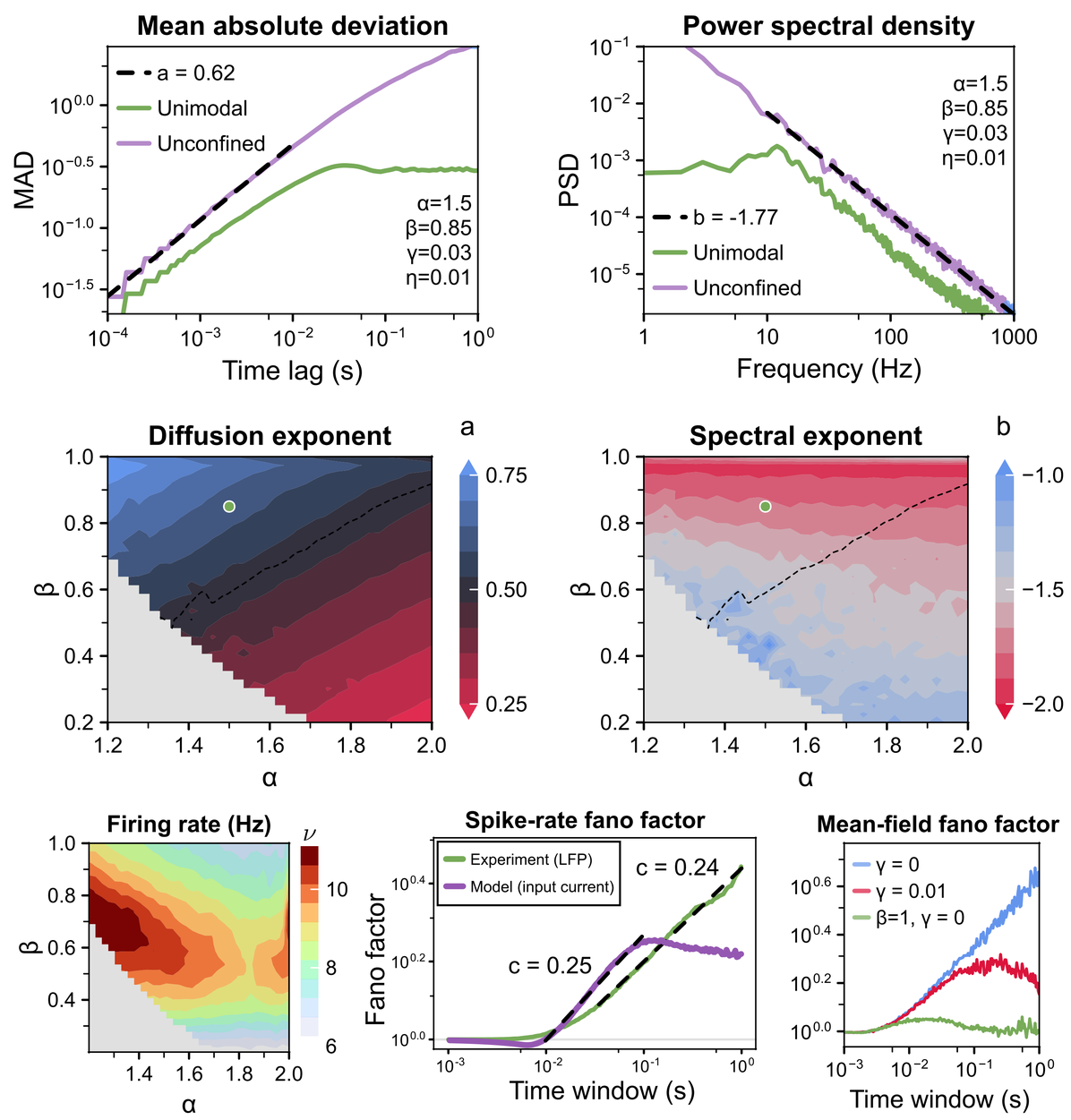
$$ {}^{C}D_t^{\beta}[f(t)] \propto \int_0^t \frac{f'(\tau)}{(t-\tau)^{\beta}}\, d\tau $$

4 | The bi-fractional model as an empirical mean field

We use the bi-fractional model as an empirical mean-field input current, allowing for **tunable** superdiffusion and long-range memory. We find:

- Superdiffusion enhances the fluctuation-driven regime
- Long-range memory gives power-law **Fano factors**, tempered by **oscillations**

The **competition** between space-fractional **superdiffusion** and time-fractional **subdiffusion** may help cortical circuits navigate the tradeoff between exploration and exploitation.



References

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